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EMDQ: Removal of Image Feature Mismatches in Real-time

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Abstract

This paper proposes a novel method for removing image feature mismatches in real-time that can handle both rigid and smooth deforming environments. Image distortion, parallax and object deformation may cause the pixel coordinates of feature matches to have non-rigid deformations, which cannot be represented using a single analytical rigid transformation. To solve this problem, we propose an algorithm based on the re-weighting and 1-point RANSAC strategy (R1P-RNSC), which operates under the assumption that a non-rigid deformation can be approximately represented by multiple rigid transformations. R1P-RNSC is fast but suffers from the drawback that local smoothing information cannot be considered, thus limiting its accuracy. To solve this problem, we propose a non-parametric algorithm based on the expectation-maximization algorithm and the dual quaternion-based representation (EMDQ). EMDQ generates dense and smooth deformation fields by interpolating among the feature matches, simultaneously removing mismatches that are inconsistent with the deformation field. It relies on the rigid transformations obtained by R1P-RNSC to improve its accuracy. The experimental results demonstrate that EMDQ has superior accuracy compared to other state-of-the-art mismatch removal methods. The ability to build correspondences for all image pixels using the dense deformation field is another contribution of this paper.

Keywords

feature matching; mismatch removal; deformation field

I. Introduction

FEATURE matching is the foundation of many computer vision applications, such as simultaneous localization and mapping (SLAM) [1] and image mosaicking [2]. Feature matching between two images typically involves multiple steps [3] [4]. The first usually entails detecting key points and extracting descriptors from the input images. This process has been studied for decades and many effective methods have been proposed, including SIFT [5], SURF [6] and ORB [7]. In recent years, deep learning-based methods have also

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been developed to improve the performance of hand-crafted features, such as LIFT [8], L2-Net [9], HardNet [10], AffNet [11], Deepdesc [12] and D2-Net [13]. Correspondences between the detected feature points can then be built according to computed descriptors using LFANN [14] or brute-force matching. However, owing to noise, incorrect matches or"mismatches" are unavoidable. Removing mismatches is essential for feature-based applications, and the related methods can be roughly classified as parametric and non-parametric methods.

Classic parametric methods assume the existence of a single analytical transformation between two sets of feature points [15]. They use the random sample consensus (RANSAC) [16] method or its variants, such as MAGSAC [17] and PROSAC [18], to estimate the transformations from small sets of randomly selected control points. The transformation models include affine [19], homography [20], epipolar geometry [21] and so on. For well-calibrated images in three-dimensional (3D) reconstruction tasks, such as SfM and SLAM, closed-form algorithms [22] can be combined with the RANSAC framework to estimate 3D camera poses and to remove mismatches simultaneously. However, single transformation-based methods cannot handle objects with non-rigid deformations. They also have limitations for rigid objects. For example, homography requires the observed objects to be approximately planar, or the translational component of the camera motion to be small. In addition, inaccurate camera calibration may lead to image distortion. Hence, even for rigid objects, it is often difficult to find a single transformation between the feature matches owing to parallax and image distortion. To solve this problem, some works have suggested the use of multiple local rigid transformations. For example, Szpak et al. proposed using multiple homographies to represent the transformation between two views of a scene consisting of multiple planar structures [23]. Puerto-Souza et al. proposed a multi-affine (HMA) algorithm [24], which uses a hierarchical k-means strategy to iteratively cluster the feature matches in small local areas and requires hundreds of milliseconds to match two images. Recently, Cavalli et al. proposed the AdaLAM algorithm that splits the image into small local regions and then assumes the correct feature matches should be consistent with the local affine transformations [25]. AdaLAM runs on the GPU in parallel to achieve real-time performance. Traditional RANSAC methods are inefficient at estimating multiple local transformations because the selected sets of control points must be (1) correct matches that (2) fit into the same rigid transformation. In this paper, we propose a novel reweighting and 1-point RANSAC-based method (R1P-RNSC) that naturally satisfies the second condition because only one control point is needed in each trial. Hence, R1P-RNSC is highly efficient compared with traditional RANSAC-based methods. R1P-RNSC was derived from our previous work [26], in which we were studying a 3D-2D correspondence problem when we discovered this strategy was even more efficient for handling the 2D-2D correspondence problem. Compared with local region-based methods [24], [25], the rigid transformations obtained by R1P-RNSC are not restricted to small local regions. Hence, it is very fast, even when running on a regular CPU.

Compared with parametric methods, non-parametric methods are more flexible in their approach to removing mismatches, especially when non-rigid deformation exists. Most non-parametric methods are explicitly or implicitly based on the assumption that the deformation is smooth in local areas, and they consider a match incorrect if it is not consistent with

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its neighbors [27] [28]. For example, Probst et al. used the isometric deformation prior, which assumes that distances between matches are preserved under deformations [29]. Ma et al. proposed a method to estimate the vector field from the matches that achieved high performance in terms of speed and accuracy [30]. They further proposed the use of machine learning methods to exploit the consensus of local neighborhood structures [31]. Hence, the representation of local smoothing information is essential to the accuracy of non-parametric methods. In this paper, we propose an algorithm called expectation-maximization and dual quaternion (EMDQ). This algorithm generates the smooth deformation field by interpolating among the feature matches, simultaneously removing mismatches according to the deformation field. Dual quaternion [32] is a mathematical tool that was introduced to the computer vision field by Newcombe et al. [33] for smooth interpolation among rigid transformations. To reduce the impact of mismatches during the interpolation, we use the expectation-maximization algorithm to adjust the weights of the feature matches.

In recent years, end-to-end deep learning-based outliers filtering methods have also been developed. For example, the Context Networks (CNe) [34] proposed a deep neural network based on PointNet [35] to distinguish correct and incorrect matches from the input coordinates of sparse feature matches, and many following works have been proposed [36] [37] [38] [39]. However, the performance of these deep learning-based outliers filtering method in practical applications is still unclear [3].

The method proposed in this paper has been designed for rigid and smooth deforming environments. The input of our method is the pixel coordinates of two sets of feature matches. As shown in Fig.1, our method consists of two steps: R1P-RNSC and EMDQ. R1P-RNSC is used to extract rigid transformations of the feature matches, and the transformation model consists of rotation, translation and scaling. The results of R1P-RNSC are essential to improving the accuracy of the interpolations derived from the subsequent EMDQ algorithm. The experimental results show that combining the two algorithms not only yields superior accuracy compared with other state-of-the-art methods, but also has sufficient speed for realtime applications. In addition, compared with other mismatch removal methods, our method can provide the deformation field for all image pixels. This approach has many potential applications, such as non-rigid image mosaicking [40].

This paper is organized as follows. In Section 2, we give the details of R1P-RNSC, including the re-weighting strategy and the termination conditions of the RANSAC framework. The details of the EMDQ algorithm are given in Section 3, in which we also provide the strategy of initialization based on the results of R1P-RNSC. Evaluation results are presented in Section 4. A discussion and description of planned future work is described in Section 5.

A. Related Works

One classic workflow of feature matching between two images is (1) detecting feature points and extracting the descriptors from the images, (2) creating a list of tentative matches by nearest-neighbor search in descriptor space, and (3) pre-filtering the incorrect feature matches [3]. For the first step, there are many effective methods to detect key points and extract the descriptors from images, such as SIFT [5], SURF [6], ORB [7], A-SIFT [41] and LIFT [8]. For the second step, there are also many works that aim to develop more robust

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and efficient matchers, such as FLANN [42] [43]. A recent work by Sarlin et al. proposed a learning-based feature matching method and has achieved better accuracy [44]. However, mismatches are often unavoidable after the first step, hence a following step to remove the mismatches is necessary. The problems in the first step have been studied for many years but less attention has been given to identifying outliers in the obtained matches.

It is very common for the correct matches in the images to have non-rigid deformations, because even when observing a rigid three-dimensional (3D) object from different angles and positions, the displacements of different parts of the object on the two-dimensional (2D) images are often non-rigid. Nonparametric methods are more flexible in handling the non-rigid deformation. For example, Li et al. proposed the ICF algorithm [27] based on a diagnostic technique and SVM to learn correspondence functions that mutually map one point set to the other. Mismatches are then identified according to their consistency with the estimated correspondence functions. Lin et al. proposed the SIM algorithm [19], which achieves affine invariance by computing the shape interaction matrices of two corresponding point sets. SIM detects the mismatches by removing the most disparate entries between the interaction matrices. Basically, these methods interpolate a non-parametric function by applying the prior condition, in which the motion field associated with the feature correspondence is slow-and-smooth. Similarly, a recent work by Lin et al. [45] proposed a regression algorithm based on as-smooth-as-possible piecewise constraints to discover coherence from the input matches.

Graph matching [46] [47] is another type of feature-matching method that uses nodes and edges to represent features and connections, respectively. For example, Leordeanu et al. proposed a spectral approach using a weighted adjacency matrix to represent pairwise agreements between matches [48]. Torresani et al. proposed a graph-matching optimization technique to minimize the energy function defined by the appearance and spatial arrangement of the features [49]. Liu et al. proposed a graph-based method to handle multiple patterns [50]. Cho et al. proposed a method based on hierarchical agglomerative clustering [51]. Although the graph-matching methods are flexible and can be used for multiple types of transformations, they suffer from their NP-hard nature.

II. Re-weighting and 1-point RANSAC (R1P-RNSC)

The input of our method is the pixel coordinates of two sets of feature matches. Under a reasonable assumption that a non-rigid deformation of the feature matches can be approximatively represented by multiple rigid transformations, we propose a method based on the re-weighting and 1-point RANSAC (R1P-RNSC) strategy to handle matching outliers when non-rigid deformation exists. Compared with traditional RANSAC methods that are based on 5- [22], 7- [52] or 8 [53] control points, the ability to use only one control point to estimate the rigid transformations can naturally satisfy the requirement that all control points should have a similar rigid transformation.

The R1P-RNSC method is fast and can provide good initial values for the subsequent refinement step. The basic idea of R1P-RNSC was derived from our previous work [26], which studies the perspective-*n*-points (P*n*P) problem. The main differences of R1P-RNSC

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and our previous PnP algorithm are as follows: (1) R1P-RNSC studies the non-rigid 2D-2D correspondence problem while the PnP algorithm studies the rigid 3D-2D correspondence problem. (2) R1P-RNSC preserves multiple rigid transformations with a modified RANSAC framework to represent the non-rigid deformation, and the PnP algorithm uses standard RANSAC.

A. Re-weighting

Denote the two coordinates of the ith match as $\mathbf{x}_i \in \mathbb{R}^2$ and $\mathbf{y}_i \in \mathbb{R}^2, i = 1, 2, \ldots N$. The rigid transformation between $\mathbf{x}_i$ and $\mathbf{y}_i$ can be represented by

$$\mathbf{y}_i = \mu(\mathbf{R}\mathbf{x}_i + \mathbf{t}), \tag{1}$$

where $\mathbf{R} \in SO(2)$ is the rotation matrix, $\mathbf{t} \in \mathbb{R}^2$ is the translation vector and $\mu \in \mathbb{R}$ is the scale factor.

In the R1P-RNSC algorithm, we randomly select a match o as the control point, then the coordinates of other matches with respect to match o are

$$\mathbf{y}_i - \mathbf{y}_o = \mu\mathbf{R}(\mathbf{x}_i - \mathbf{x}_o), i = 1, 2, \ldots, N, \tag{2}$$

which suggests that a rigid transform can be represented by $\mathbf{R}$, μ, $\mathbf{x}_o$ and $\mathbf{y}_o$. Because $\mathbf{x}_o$ and $\mathbf{y}_o$ are constants, only $\mathbf{R}$ and μ are needed to be estimated. According to Ref. [54], $\mathbf{R}$ and μ can be obtained from $\mathbf{x}_i$ and $\mathbf{y}_i$, $i = 1, 2, \ldots N$, by

$$\left[\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}^T\right] = \mathrm{svd}\left(\mathbf{Y}\mathbf{X}^T\right), \mathbf{R} = \mathbf{U}\mathbf{V}^T, \tag{3}$$

$$\mu = \| \mathrm{vector}(\mathbf{Y}) \| / \| \mathrm{vector}(\mathbf{X}) \|, \tag{4}$$

where

$$\mathbf{X} = \left[\mathbf{x}_1 - \mathbf{x}_o \cdots \mathbf{x}_N - \mathbf{x}_o\right]_{2 \times N}, \tag{5}$$

$$\mathbf{Y} = \left[\mathbf{y}_1 - \mathbf{y}_o \cdots \mathbf{y}_N - \mathbf{y}_o\right]_{2 \times N}. \tag{6}$$

Because incorrect matches exist and not all correct matches can fit into the rigid transform of match o, the estimation of the rigid transformation represented by $\mathbf{R}$ and μ may be incorrect. We propose a re-weighting method to dynamically update the weights of matches $i = 1, 2, \ldots N$ by

$$d_i = \|\mathbf{y}_i - \mathbf{y}_o - \mu\mathbf{R}(\mathbf{x}_i - \mathbf{x}_o)\|, \tag{7}$$

$$w_i = \min(H/d_i, 1), \tag{8}$$

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where d_i is the error distance of match i with the estimated **R** and μ, H is a pre-defined threshold that when the error distance $d_i < H$, match i is recognized as correct.

With the obtained weights of matches w_i, $i = 1, 2, \ldots N$, the related items of matrices **X** and **Y** are adjusted according to the weights. Small weights w_i will be assigned to both incorrect matches and correct matches that cannot fit into the same rigid transformation of match o, hence the estimation of **R** and μ in (3) and (4) will not be affected.

We perform the re-weighting strategy within an iterative process. With a randomly selected match o as the control point, we set the initial values of weights as w_i for $i = 1, 2, \ldots N$. Then, we alternatively update **R** and μ according to (3) and (4), and update the weights of matches w_i according to (7) and (8). In practice we find this iteration process only requires a very small number of iterations to converge, and we choose the number of iterations 3 in the experiments.

At the end of the iteration process, we recover the translation **t** by

$$\mathbf{t} = \mathbf{y}_o/\mu - \mathbf{R}\mathbf{x}_o, \tag{9}$$

which is not needed by R1P-RNSC for removing incorrect matches, but the subsequent EMDQ algorithm requires the estimation results of rigid transformations represented by **R**, **t** and μ to generate the initial dual quaternions.

B. Modified RANSAC

Traditional RANSAC methods aim to find the consensus that is supported by the most number of input data, and other consensuses are simply abandoned. It is difficult to use a single parametric transformation to represent the non-rigid deformation. To solve this problem, we propose to use multiple rigid transformations. In the R1P-RNSC algorithm, the rigid transformations are obtained in different RANSAC trials with different control points o. Hence, it is necessary to modify the standard RANSAC methods to reserve the obtained rigid transformations and design a new termination condition.

1) Results Reservation: Within the RANSAC framework, we randomly try different matches as the control point o. With a control point o, some matches may have small error distances d_i as in (7) and can be considered as the candidate inliers. Denote the number of candidate inliers detected with control point o as T_o, we omit the results if $T_o < T_{\min}$, where $T_{\min} \in \mathbb{R}$ is a threshold. A small T_o suggests that the selected match o may be incorrect and the results obtained from should be omitted. If $T_o \geq T_{\min}$, we add the candidate inliers to the final detection results, and update the inliers ratio by

$$\gamma = N_{\text{inliers}}/N, \tag{10}$$

where N_{inliers} is the total number of detected inliers.

In our algorithm, the control point o is only selected from the matches that are not considered correct so far, which aims to avoid repeatedly find the same rigid transformation.

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2) RANSAC Termination: The standard termination condition of RANSAC is determined by the number of trials and the largest number of detected inliers in one trial, which needs to be modified for the R1P-RNSC algorithm. The basic idea of the modified RANSAC termination condition is to terminate the RANSAC process when inliers are unlikely to exist in the remaining matches.

T_{min} is the minimal number of inliers detected by one trial that can be reserved. Assuming that there exists a set of T_{min} inliers in the remaining matches, then the possibility that a selected control point o belongs to this set of inliers is $T_{\text{min}}/(N - \gamma N)$. Then, the possibility that after k trials of different control point o, the possibility that a set of T_{min} inliers is not found is $(1 - T_{\text{min}}/(N - \gamma N))^k$, which equals to $1 - p$. p is the widely used RANSAC parameter [55], which suggests the desired probability that the RANSAC algorithm provides a useful result after running and in our algorithm we set $p = 0.95$. Hence, the termination of our 1-point RANSAC framework is

$$k > \frac{\log(1-p)}{\log(1 - T_{\text{min}}/(N - \gamma N))}, \tag{11}$$

where k is the number of RANSAC trials.

C. Sparse Implementation

The R1P-RNSC algorithm computes the rigid transformations by re-weighting all matches. A more efficient way is to use a small number of sample points $N_{\text{sparse}} < N$ to compute the rigid transformations, and then apply the obtained rigid transformations back to all matches to determine inliers and outliers. This sparse implementation is less accurate because it may miss possible rigid transformations to represent the non-rigid deformation, but is faster due the the smaller number of matches involved in the re-weighting process.

D. Discussion of R1P-RNSC

The proposed R1P-RNSC algorithm is a parametric method that is efficient in estimating rigid transformations from the matches when outliers exist. It is worth nothing that R1P-RNSC does not consider the distances between feature matches. Hence, the obtained rigid transformations are not restricted to small local areas. For example, if the transformation between the two images is rigid, R1P-RNSC will only need one RANSAC trail to cover all feature matches. Compared with the multi-affine method [24] that splits the images into small local areas, our R1P-RNSC method can take into account all matches at the same time and is hundreds of times faster. However, the neglect of local smoothing information limits the accuracy of R1P-RNSC. To solve this problem, we propose a non-parametric algorithm called as EMDQ as the refinement step, which generates the smooth deformation field by interpolating among the correct matches, and simultaneously remove incorrect matches according to the deformation field.

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Algorithm 1 :

Pseudo code of R1P-RNSC

Input: N coordinates of matches $x_1, \ldots, x_N$ and $y_1, \ldots, y_N$

Initialization:

label$_i$ = False, $d_i' = 1e + 8$ for $i = 1, 2, ..N$

$punish = 1e + 8$

while $k < 500$ **do**

$k = k + 1$

if $k >$ *punish* **then** break; ⊳ equivalent to Eq. (11)

randomly select $o \in (1, N)$

compute **X** and **Y** by Eq. (5) and (6)

$\mathbf{R} = \mathbf{I}_{2\times 2}, \mu = 1$

for iter = 1:3 **do**

compute d_i and w_i by Eq.(7) and (8)

compute $\mathbf{X}'$ and $\mathbf{Y}'$ by multiplying w_i to related columns of **X** and **Y** respectively

compute **R** and μ from $\mathbf{X}'$ and $\mathbf{Y}'$ by Eq. (3) and (4)

end for

if $d_i < H$ **then**

label$_i$ = True

end if

$T_o = \sum(\text{label}_i)$

if $T_o < T_{\min}$ **then** continue;

compute γ and *punish* by Eq. (10) and Eq. (11)

compute **t** by Eq. (9)

compute dual quaternion $\mathbf{q}_i$ from **R**, **t** for match i, if label$_i$ = True and $d_i < d_i'$

$d_i' = \min(d_i, d_i')$ if label$_i$ = True

end while

Output: label$_i$, μ_i and $\mathbf{q}_i$ for $i = 1, 2, \ldots N$

III. EXPECTATION MAXIMIZATION AND DUAL QUATERNION (EMDQ)

We propose a non-parametric algorithm called as EMDQ to remove incorrect matches by generating the smooth deformation field. Denote a deformation field as f, $f(x_i)$ is the displacement of x_i caused by f. Correct matches should be consistent with the deformation field f, which suggest

$$y_i \approx f(x_i) \tag{12}$$

if match i is correct.

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According to (12), the proposed EMDQ algorithm distinguishes correct and incorrect matches according to the error distances between $\mathbf{y}_i$ and $f(\mathrm{x}_i)$, $i = 1, 2, \ldots N$. The ability to generate the smooth deformation field f from a mixed set of correct and incorrect matches is essential, and we employ dual quaternion to solve this problem.

A. Dual Quaternion-based Deformation Field

We assign discrete transformations g_i, $i = 1, 2, \ldots N$ to all matches, and the smooth deformation field f is generated by interpolating among g_i. According to (1), a transformation g_i consists of the rotation $\mathbf{R}_i$, the translation $\mathbf{t}_i$ and the scale μ_i, which moves coordinate $\mathbf{x}_i$ to $\mathbf{y}_i$ by

$$\mathbf{y}_i = g_i(\mathbf{x}_i) = \mu_i(\mathbf{R}_i\mathbf{x}_i + \mathbf{t}_i)\,. \tag{13}$$

Then, the value of f at a location is equal to the weighted average value of neighboring g_i. In this mismatches removal problem, we mainly focus on the values of f at the locations of the matches $\mathbf{x}_i$, $i = 1, 2, \ldots N$, that is

$$f_{\mathbf{x}_i} = \sum_{j=1}^{N} w_j^i g_j, \tag{14}$$

where w_j^i is the weight that suggests the impact from match j to i. $f_{\mathbf{x}_i}$ is the value of f at the coordinate $\mathbf{x}_i$. This weighted average strategy (14) suggests that only when the transformations of neighboring matches are consistent with that of a match i, then match i can be recognized as correct according to (12).

However, when performing interpolation according to (14), the use of the rotation matrices and translation vectors as in (13) may lead to inaccurate results [56]. To overcome this difficulty, we use a mathematics tool called as dual quaternion (DQ) [32] for smooth interpolation.

A DQ is a 8- dimension vector that can represent a 3D rotation and a 3D translation simultaneously. A 2D transform can be considered as a 3D transform that is restricted at the $z = 0$ plane, and 4 out of the 8 components of a DQ is always zero. To improve the efficiency, we only compute the 4 nonzero components in our EMDQ implementation. Denote a dual quaternion related to match i as $\mathbf{q}_i$, we have

$$\mathbf{y}_i = g_i(\mathbf{x}_i) = \mu_i\mathbf{q}_i(\mathbf{x}_i), \tag{15}$$

where $\mathrm{q}_i(\mathrm{x}_i)$ suggests to apply $\mathbf{q}_i$ to the coordinate $\mathrm{x}_i \in \mathbb{R}^D$ to obtain a transformed coordinate.

According to (14) and (15), the deformation field f at the location $\mathbf{x}_i$ is determined by the weighted average value of μ_j and $\mathbf{q}_j$, that is

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$$f_{\mathbf{x}_i} = \bar{\mu}_i \bar{\mathbf{q}}_i \tag{16}$$

Where

$$\bar{\mathbf{q}}_i = \sum_{j=1}^{N} \left(w_j^i \mathbf{q}_j\right) / \left\| \sum_{j=1}^{N} \left(w_j^i \mathbf{q}_j\right) \right\|_{dq}, \tag{17}$$

$$\bar{\mu}_i = \sum_{j=1}^{N} \left(w_j^i \mu_j\right) / \sum_{j=1}^{N} \left(w_j^i\right), \tag{18}$$

where $\| \cdot \|_{dq}$ is the norm of the dual quaternion.

Many works in the computer graphics field have proven the feasibility and efficiency of using the linear combination of DQ as in (17) to generate smooth interpolations. For example, by using DQ-based interpolation from skeletons, the motion of the skin surface of a mesh model can be generated [32]. However, due to the existence of incorrect matches, the interpolation process (17) and (18) may be incorrect. One way to reduce the impacts of outliers is to assign a small weight w_j^i when match j is incorrect. Hence, we employ the expectation maximization (EM) algorithm to dynamically update the weights w_j^i in the EM iteration process.

B. EM Algorithm

The EM algorithm is an iterative process that alternatively performs two steps. In the E-step, the probabilities of matches being correct, or ”inliers”, are computed, and the weights w_j^i are further updated. In the M-step, the deformation field f is updated according to the weights w_j^i. The details of the EM algorithm is as follows.

(1) E-step: Inspired by [57], we consider the mismatches removal problem as a classification problem according to the error distances between $\mathbf{y}_i$ and $f(\mathrm{x}_i)$, $i = 1, 2, \ldots N$. With a deformation field f, we have

$$p(i \mid \text{inlier}) = \frac{1}{2\pi\sigma^2} \exp\left(\frac{\left\| \mathbf{y}_i - f(\mathbf{x}_i) \right\|^2}{2\sigma^2} \right) \tag{19}$$

where σ is the standard deviation.

We consider the incorrect matches, or ”outliers”, follow an uniform distribution, that is

$$p(i|\text{outlier}) = a, \tag{20}$$

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where a is a constant.

Denote γ as the inliers ratio, we have $p(\text{inlier}) = \gamma$ and $p(\text{outlier}) = 1 - \gamma$. According to Bayes' theorem, we have

$$\begin{aligned} & p(\text{inlier} \mid i) \\ &= \frac{p(i \mid \text{inlier}) p(\text{inlier})}{p(i \mid \text{inlier}) p(\text{inlier}) + p(i \mid \text{outlier}) p(\text{outlier})} \\ &= \frac{\exp\left(\left\|\mathbf{y}_i - f(\mathbf{x}_i)\right\|^2 / 2\sigma^2\right)}{\exp\left(\left\|\mathbf{y}_i - f(\mathbf{x}_i)\right\|^2 / 2\sigma^2\right) + 2\pi\sigma^2 \frac{(1-\gamma)}{\gamma} a} \end{aligned} \tag{21}$$

$p(\text{inlier} \mid i)$ suggests the probability that match i is an inlier. We adjust the weight w_j^i, which is the influence of match j to match i, according to $p(\text{inlier} \mid j)$. Taking into account the distances between matches, w_j^i is determined by two parts, that is

$$w_j^i = w_{j,\,\text{distance}}^i p(\text{inliers} \mid j), \tag{22}$$

where $w_{j,\,\text{distance}}^i$ is determined by the distance between match i and j, that is

$$w_{j,\,\text{distance}}^i = \max\left(\exp\left(\frac{\left\|\mathbf{y}_i - \mathbf{y}_j\right\|^2}{2r^2}\right), \exp\left(\frac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{2r^2}\right)\right) \tag{23}$$

Where $r \in \mathbb{R}$ is a pre-defined radius. Because $w_{j,\,\text{distance}}^i$ does not contain variables, we only compute it at the initial phase. Eq. (23) treats the two images equally, which can improve the accuracy when computing the local smooth deformation field because the distances between feature matches in both images can be taken into account.

(2) M-step: According to the updated weight w_j^i, we update $\bar{\mathbf{q}}_i$ and $\bar{\mu}_i$ for each match i according to (17) and (18) respectively, which represent the value of the deformation field f at $\mathbf{x}_i$, or $f\mathbf{x}_i$.

The standard deviation σ is updated by

$$\sigma^2 = \sum_{i=1}^{N} \left(p(\text{inlier} \mid i)\left\|\mathbf{y}_i - f(\mathbf{x}_i)\right\|^2\right) / \sum_{i=1}^{N} p(\text{inlier} \mid i)\,. \tag{24}$$

Finally, we update q_i to keep the transformation g_i close to the deformation field $f\mathbf{x}_i$ by

$$\Delta\mathbf{q}_i = trans2\mathrm{dq}((\mathbf{y}_i - f(\mathbf{x}_i))/\bar{\mu}_i), \tag{25}$$

$$\mathbf{q}_{i,\,\text{new}} = \mathbf{q}_{i,\,\text{old}}\, \Delta\mathbf{q}_i\,. \tag{26}$$

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Equations (25)-(26) aim to keep (15) holding and make the transforms g_i more smooth, and trans2dq represents the conversion of the transform to the DQ representation.

The termination condition of the EM algorithm determined by the change $p(\text{inlier}|i)$, that is

$$\frac{1}{N}\left(\sum_{i=1}^{N} \mid p(\text{inlier} \mid i)^{(k)} - p(\text{inlier} \mid i)^{(k-1)} \mid\right) < \theta, \tag{27}$$

where θ is a threshold. k suggests the value of the kth iteration in the EM algorithm. After termination, matches with

$$p(\text{inlier} \mid i) > p_{\min} \text{ and } \mathbf{y}_i - f(\mathbf{x}_i) < H \tag{28}$$

are considered as inliers, where $p_{\min}$ is a threshold and H is the same threshold used in the R1P-RNSC algorithm.

Algorithm 2 :

Pseudo code of EMDQ

Input: N coordinates of matches $\mathrm{x}_1, \ldots, \mathrm{x}_N$ and $\mathrm{y}_1, \ldots, \mathrm{y}_N$, μ_i and $\mathbf{q}_i$ from R1P-RNSC

Initialization:

set $w^i_{j,\,\text{distance}}$, by Eq. (23),

set $\sigma^2 = H^2$,

set inliers ratio γ from the R1P-RNSC result, $\gamma \in [0.05, 0.95]$

set $p(\text{inlier}|i)$ by Eq.(29)

while $k < 20$ **do**

E-step:

compute w^i_j by Eq. (22)

compute f by Eq. (16)

compute $p(i|\text{inlier})$ by Eq. (19)

compute $p(\text{inlier}|i)$ by Eq. (21)

M-step:

compute σ^2 by Eq. (24)

compute the inliers ratio γ by Eq. (28), $\gamma \in [0.05, 0.95]$

compute $\mathbf{q}_i$ by Eq. (26)

if Eq. (27) is True **then**

break;

end if

$k = k + 1$

end while

get label_i by Eq. (28)

Output: label_i, μ_i and $\mathbf{q}_i$ for $i = 1, 2, \ldots N$

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C. Initialization from R1P-RNSC Results

There are mainly three variables in the EMDQ algorithm that need to be initialized, which include the dual quaternions $\mathbf{q}_i$, the scale factors μ_i and the weights between matches w_j^i. According to (22), w_j^i is determined by $w_{j,\text{distance}}^i$ and $p(\text{inliers} \mid j)$, and $w_{j,\text{distance}}^i$ is constant in the EM iterations. The initial values of the three variables determine the initial deformation field f according to (14), and we use the results of R1P-RNSC for the initialization.

The proposed R1P-RNSC algorithm detects multiple rigid transformations. If a match i is considered as an inlier with a rigid transformation detected by R1P-RNSC, then the related $\mathbf{q}_i$ and μ_i will be initialized according to the results. A match i may satisfy multiple rigid transformations, and we use the one that has the least error for EMDQ initialization. The rigid transformations obtained by R1P-RNSC can improve the accuracy of the interpolation, as shown in Fig. 2 and 3.

The initial values of $p(\text{inliers} \mid i)$ are important for reducing the impact of outliers when generating the initial deformation field f. The natural idea is to initialize $p(\text{inliers} \mid i) = 1$ if match i is recognized as an inlier by R1P-RNSC, or $p(\text{inliers} \mid i) = 0$. However, because at some local areas, there may be more false positives than true positives in the R1PRNSC results, and the generated deformation field f yields false positives. We propose a simple method to overcome this problem according to the uncertainty of the detected rigid transformations provided by R1P-RNSC. With match o as the control point, R1P-RNSC may find different number of inliers, which is denoted as T_o. It is intuitive that when T_o is large, the results are more reliable. Hence, we initialize

$$p(\text{inliers} \mid i) = T_o / \sum T_o(i) \tag{29}$$

if match i is recognized as an inlier with match o as the control point. This allows that f is initially generated from the most reliable matches.

IV. EXPERIMENTS

The performance of the proposed EMDQ algorithm was evaluated against the state-of-the-art mismatch removal methods. The source code was implemented in MATLAB and executed on a computer with an Intel Core i7 2.60 GHz CPU. The Matlab and C++ source code are available at https://github.com/haoyinzhou/EMDQ and https://github.com/haoyinzhou/EMDQ_C respectively. The Matlab code also includes the other methods that are used for comparison in this paper. In our MATLAB implementation, we used the vlfeat open source library [58] for generating the kd-tree to obtain the neighboring matches.

The parameters are as follows: $H = 20$ pixels is the inliers threshold; $T_{\min} = 5$ is the least number of correct matches found by one R1P-RNSC trial that will be added to the final R1P-RNSC results; $r = 80$ pixels is the radius to compute $w_{j,\text{distance}}^i$ between matches; $a = 1e-5$ is used for computing $p(i \mid \text{outlier})$; $p_{\min} = 0.5$ is the threshold for EMDQ to

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determine inliers and outliers; $\theta = 0.005$ is threshold used in the termination condition of EMDQ iterations; $N_{\text{neighbor}} = 20$ is the number of neighboring matches that are considered in the EMDQ algorithm, which are found by kd-tree. $N_{\text{sparse}} = \min(100, N)$. The above parameters are not sensitive. For example, in practice we found that using a significantly different parameters $H = 100$ does not change the results in most situations. However, when the deformation or distortion between two images are large, we suggest to use a larger H to make R1P-RNSC can tolerate larger differences between the estimated rigid transformations and the real non-rigid deformation.

The EMDQ algorithm has been designed for rigid or smooth deforming environments. Hence, the experiments were conducted on the two situations.

A. Rigid Environment

The rigid environment experiments were mainly conducted on DTU robot image data [1] [59], which provides images and the related 3D point cloud obtained by the structured light scan. The camera was fully calibrated and the true values of the camera positions and orientations are known. Images have a resolution of 800×600. Datasets numbered 1 to 30 were used. In each dataset, the images were captured under 19 different illumination situations and from 119 camera positions. We randomly selected 5 out of 19 illumination situations. Hence, a total of 30 × 5 × 119 = 17850 images were included in this evaluation. Following instruction of the DTU dataset, for each dataset and illumination situation, we used image number 25 as the reference image and performed SURF matching between the reference image and other images, which were captured within approximately ±36 degrees of the reference image. A larger camera shot angle between two images often leads to a larger percentage of incorrect matches. By reprojecting the 3D point clouds to the images according to the true values of the camera positions and orientations, we designated matches with a re-projection error smaller than 10 pixels as inliers.

The EMDQ algorithm requires R1P-RNSC for initialization. Hence, in this section, EMDQ reports the final accuracy and runtime results of the R1P-RNSC and EMDQ algorithms combined, whereas the sEMDQ reports the results when using the sparse implementation of R1P-RNSC. We also provide the results of EMDQ without using R1P-RNSC initialization, which is denoted as EMDQ (no RNSC). EMDQ (no RNSC) was initialized under the assumption that displacements of the feature matches were caused by translational motions only, which suggests $\mu = 1$ and $\mathbf{R} = I$ for all matches.

The mismatch removal methods used for comparison included AHC [20], LPM [60], LMR [31], SIM [19], VFC [30], Context Network (CNe) [34] and OANet [61]. All these methods have been recently proposed and thus reflect state-of-the-art performance. Among these methods, CNe, OANet and LMR are deep learning-based mismatch filtering methods, and the others are not learning-based. For our experiments, we used the pre-trained models provided by the authors of the machine learning-based methods. We have also included the traditional RANSAC + homography and RANSAC + epipolar geometry methods [2].

[1] http://roboimagedata.compute.dtu.dk/
[2] https://www.robots.ox.ac.uk/vgg/hzbook/code/

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VFC has a fast version and a sparse version, referred to as fVFC and sVFC, respectively. AHC is a very fast method for removing matching outliers and focuses on 3 × 3 projective transformations, namely, planar homographies. However, for the DTU dataset, the images were captured with rotating 3D objects; hence, the planar homographies assumption may not stand.

As shown in Fig.4, the total number of matches and the percentage of outliers varied with objects, illumination situations and camera pose. Although clear comparisons require only one factor be different, this kind of variable-controlling is difficult when evaluating real-world data because the SURF matching results are unpredictable. In experiments, we found that the accuracy of the outlier removal algorithms was mainly affected by the percentage of outliers, and the runtime was mainly affected by the total number of matches. Therefore, in this section, we report the accuracy and runtime evaluation results by comparing the statistical results for each percentage of outliers and the range of total number of matches, respectively.

For accuracy evaluation, we report four metrics, which include the number of errors, *F-score*, *recall* and *precision*. The number of errors suggests the total number of differences between the ground truth and the inliers-outliers recognition results. *F-score* is widely used for statistical analysis of binary classification, and we use the balanced *F-score*

$$F\text{-}score = 2\,Recall \cdot Precision\,/(Recall + Precision) \tag{30}$$

for the DTU experiments. We are aware that the *Precision* rate is relatively more important for the mismatches removal task, and there exist *F-score* that can weight *Precision* higher than *recall*. However, our experimental results show that our methods have better *Precision* rates than most other methods. Hence, the balanced *F-score* in Eq. (30) may reflect the performance of our method better.

As shown in Fig. 5 and 7, the proposed EMDQ method has the highest accuracy in terms of number of errors and the *F-score*. sEMDQ did not show an obvious decrease in accuracy compared with EMDQ. fVFC and sVFC yielded very similar results; hence, we combined them in Fig. 5. When the inliers ratio was high, the *recall* rate of fVFC/sVFC was lower than EMDQ/sEMDQ but the *precision* rate was slightly higher. As the inliers ratio decreased, the *precision* rate of fVFC/sVFC dropped significantly, which decreased the *F-score* significantly although the *recall* rate remained high. When we compared EMDQ and EMDQ (no RNSC), it became clear that R1P-RNSC initialization improves the accuracy of EMDQ. Fig. 5 shows that EMDQ (no RNSC) has a similar accuracy with EMDQ when the percentage of outliers is low because a low percentage of outliers often suggests a small change of camera pose between the two images. Hence, the translational motion-based initialization is similar to the R1PRNSC initialization. The direct outputs of Context Network (CNe) and OANet are the scores for each pair of matches that suggest the binary classification results, and we report the results with different thresholds in Fig. 5. When using a strict threshold $T = 0$ or $T = 1$, CNe and OANet only considered a few matches to be correct. Therefore, its *precision* rate is high but the *recall* rate is much lower than other methods. In our experiments, we found the threshold $T = -2.5$ yielded a balanced trade-off

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between *precision* and *recall* for CNe and OANet; however, the number of errors and the *F-score* were not very satisfying.

As shown in Fig. 6, sEMDQ had a comparable runtime with sVFC. AHC was fastest because it assumes that the object is planar, namely the planar homographies problem. This assumption makes AHC not suitable for solving the general outliers removal problem. Hence as shown in Fig. 5, the accuracy of AHC was lower than EMDQ and VFC. EMDQ were among the most performant methods in terms of speed.

Another quantitative experiment was conducted on the VGG dataset [3], as shown in Fig. 8. The data include planar rigid objects, and the ground-truth homography matrices are given. In this experiment, we used the AffNet [11] method to detect feature points and extract descriptors, and matches were built according to the AffNet descriptors using brute-force searching and the nearest neighbor ratio (NNR) (threshold=0.8). AffNet finds many more matches than SURF especially when the two images have significant orientations. The quantitative results are given in Table I. The proposed EMDQ algorithm provides relatively accurate results in terms of the number of errors and $F-score$. However, since EMDQ aims to handle more general situations, such as smooth deforming environments, it is not as accurate as homography-based methods, such as AHC and the traditional RANSAC+homography method. Conversely, the homography-based methods are not as accurate in handling general 3D situations, as shown in the DTU experiments in Fig. 5. Note: in these experiments, we use the following $F\,score$, which weighs *recall* lower than *precision*:

$$Fscore = \left(1 + 0.5^2\right)\frac{precision * recall}{0.5^2\, precision + recall}. \tag{31}$$

In addition to the quantitative experiments, we also conducted qualitative experiments. As shown in Fig. 9, the images were obtained from the sun3D dataset [4], we used AffNet to obtain matches and the generated deformation fields by EMDQ are smooth and physically realistic. As shown in Fig. 10, another experiment was conducted on natural images, which were usually used in the image mosaicking task. The third qualitative experiments was conducted on fisheye images as shown in Fig. 11. Fisheye images have extremely large image distortion, and the ability to handle fisheye images show that our method can handle un-calibrated images.

B. Smooth Deforming Environments

To demonstrate the ability to handle smooth deforming non-rigid environments, we conducted qualitative experiments on the Mandala dataset [62], which include videos obtained by scanning objects with large and fast deformation. As shown in Fig. 12 and 13, the results of the EMDQ algorithm are visually accurate. As shown in our recent paper [40], we can compensate for the deformation of the object and mosaic the images

[3] https://www.robots.ox.ac.uk/vgg/data/affine/
[4] http://sun3d.cs.princeton.edu/

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according to the deformation field by using the EMDQ algorithm as a black box. Since the prerequisite of non-rigid image mosaicking is to remove mismatches and obtain the non-rigid transformations between images, this experiment demonstrates that EMDQ can handle non-rigid objects robustly.

V. CONCLUSION

In this paper, we propose a novel EMDQ algorithm to address the mismatch removal problem, which serves as the subsequent step of the feature-matching method. The inputs of the EMDQ algorithm are the pixel coordinates of the feature matches, and the outputs include (1) the binary labels that distinguish correct and incorrect feature matches, and (2) the dense deformation field that provides registration for all pixels between the two images. Experimental results show that EMDQ is superior for mismatch removal compared with other state-of-the-art mismatch removal methods.

Although the major goal of the proposed EMDQ algorithm is to address the mismatch removal problem, the algorithm can also generate the dense deformation field from sparse and noisy matches. Thus, EMDQ can potentially be applied to problems such as image registration and mosaicking. In this paper, we have demonstrated the results of a real-time non-rigid image mosaicking method [40], which was developed based on the deformation field obtained by the EMDQ algorithm. The accuracy of the deformation field is strongly dependent on the accuracy of mismatch removal, since the two outputs are correlated in the EMDQ algorithm. EMDQ can also handle 3D cases and we refer to the arXiv version of this paper in Ref. [63].

Limitations:

EMDQ works under the assumption that the deformation field of the feature matches is smooth; hence, it cannot handle non-smooth deformation. For example, for situations involving multiple objects with different motions in the images, the EMDQ algorithm and other existing mismatch removal methods introduced in this paper may not work robustly.

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Biographies



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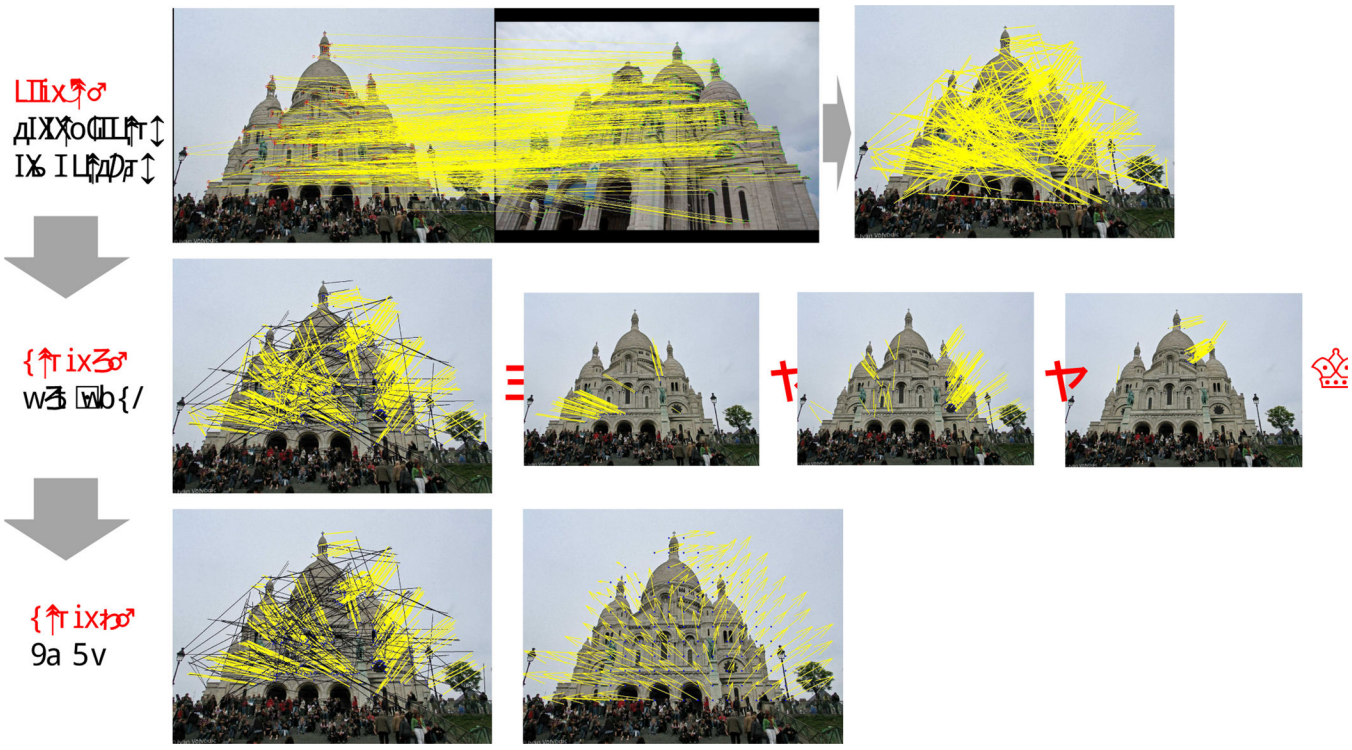


Fig. 1.

Outline of the proposed method to remove incorrect matches, or"mismatches". First row: The input images. The feature points were detected by AffNet [11], and the matches were built according to the AffNet descriptors using brute-force searching and the nearest neighbor ratio (NNR). Although the objects are rigid, the pixel displacements of the correct matches may be non-rigid owing to parallax. Second row: R1P-RNSC results assume that the non-rigid deformation can be represented by multiple rigid transformations. Third row: EMDQ results and the generated deformation field. In this example, it took approx. 20 ms to remove the outliers plus an additional 3 ms to generate the deformation field on the grid points.

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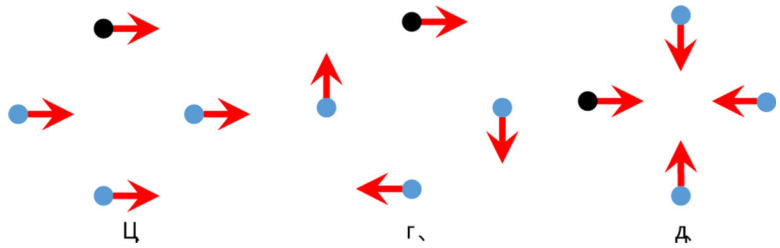


Fig. 2.

As shown in this toy model, the estimation of rigid transformations by R1P-RNSC can improve the accuracy of deformation field generation, which is essential for the subsequent EMDQ algorithm. The black points in (a)-(c) have the same displacements, which are caused by translation, rotation and scaling, respectively, according to the neighboring blue points. Hence, when performing an interpolation, the effects of the black points on the neighboring areas should be different for each of the three cases.

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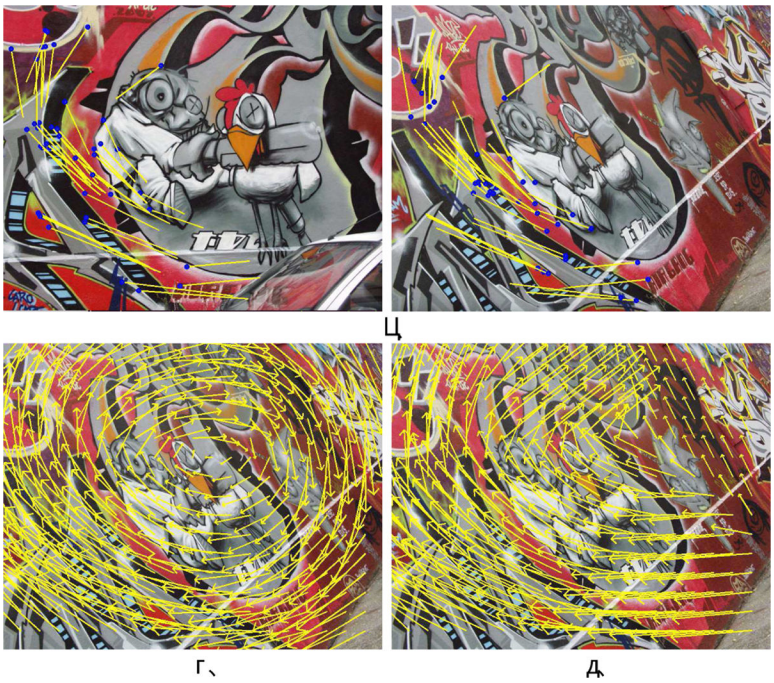


Fig. 3.

(a) Two images showing the correct SURF matches between them. (b) The deformation field obtained by interpolating among the dual quaternions of the SURF matches according to the R1P-RNSC results. (c) The resultant deformation field without using R1P-RNSC results. This field considers the displacement of the feature matches caused by translational motions only (Fig. 2). The deformation field in (b) is more accurate. Since EMDQ uses the deformation field for mismatches removal, the R1P-RNSC results can also improve the accuracy of mismatches removal.

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Fig. 4.

Samples of the results of EMDQ on the DTU images with SURF matches. First and second rows: two images are shown with SURF matches. The yellow and red lines have been identified as inliers and outliers, respectively, by EMDQ. Third row: the generated deformation field on grid sample points, which are obtained by taking the weighted average of neighboring $\mathbf{q}$ and μ of the match inliers. The deformation field is shown only at locations close to the match inliers, suggesting lower uncertainty.

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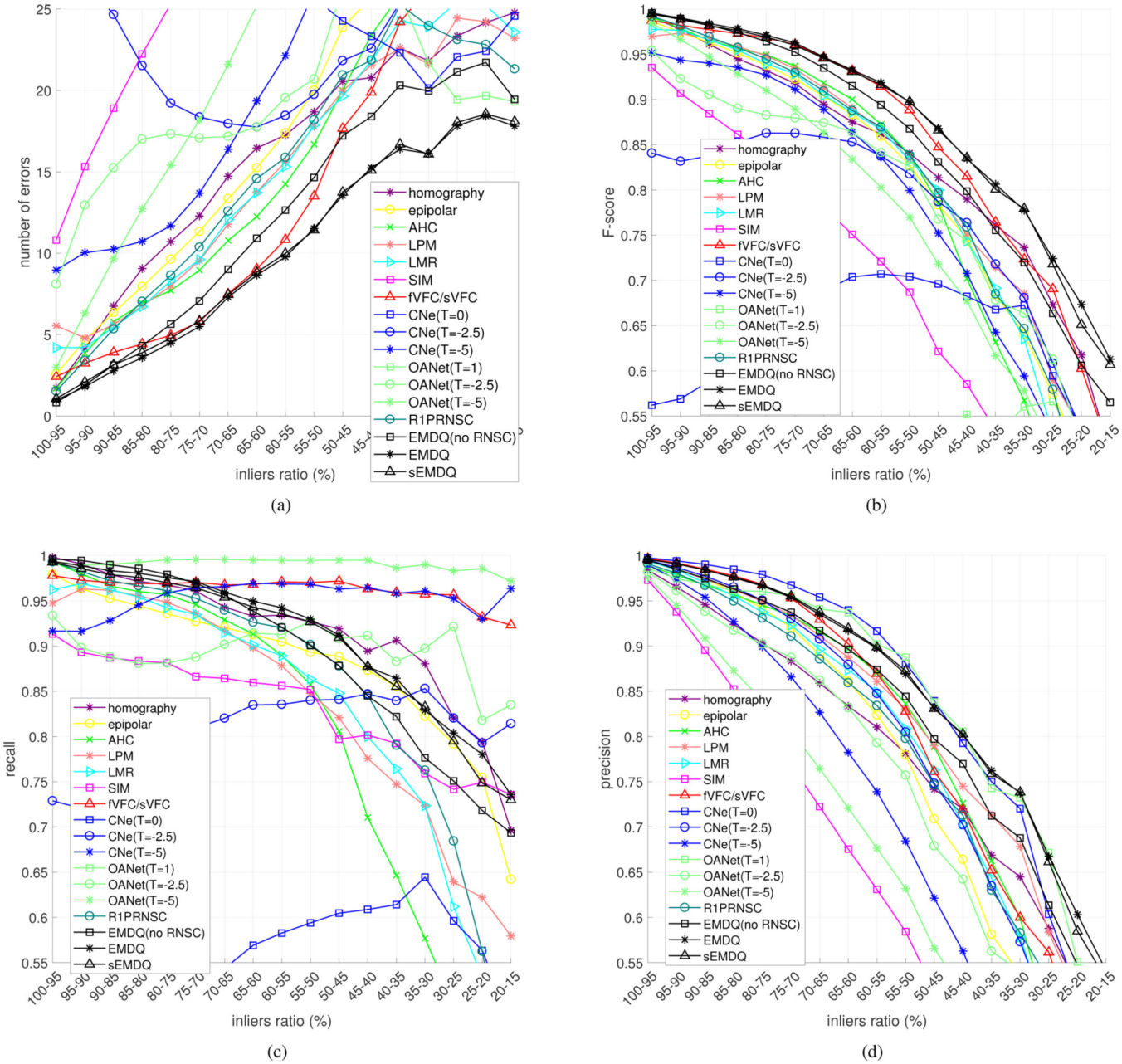


Fig. 5.

Accuracy results on the DTU data. (a) Number of errors. (b) *F-score*. (c) *Recall*. (d) *Precision*.

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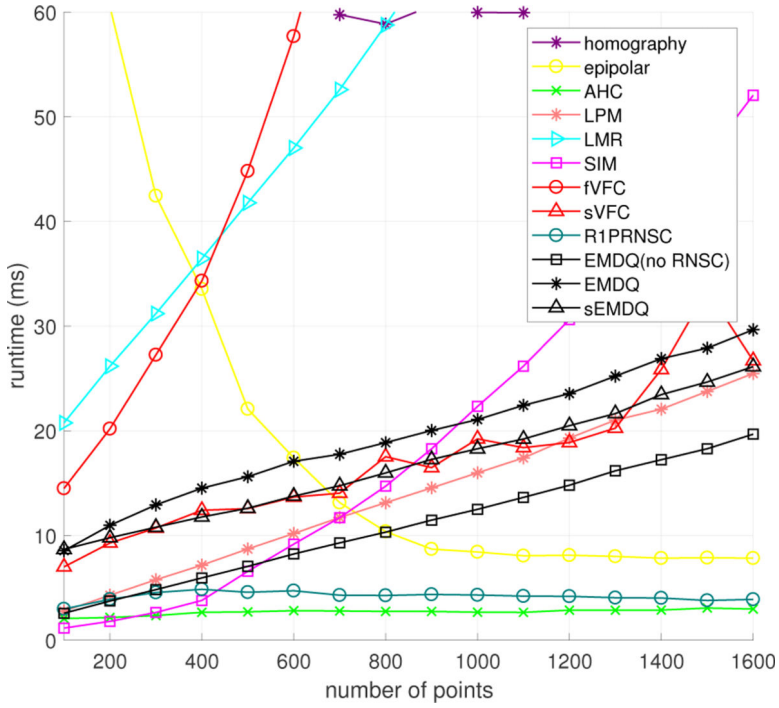


Fig. 6.
Runtime results on the DTU data.

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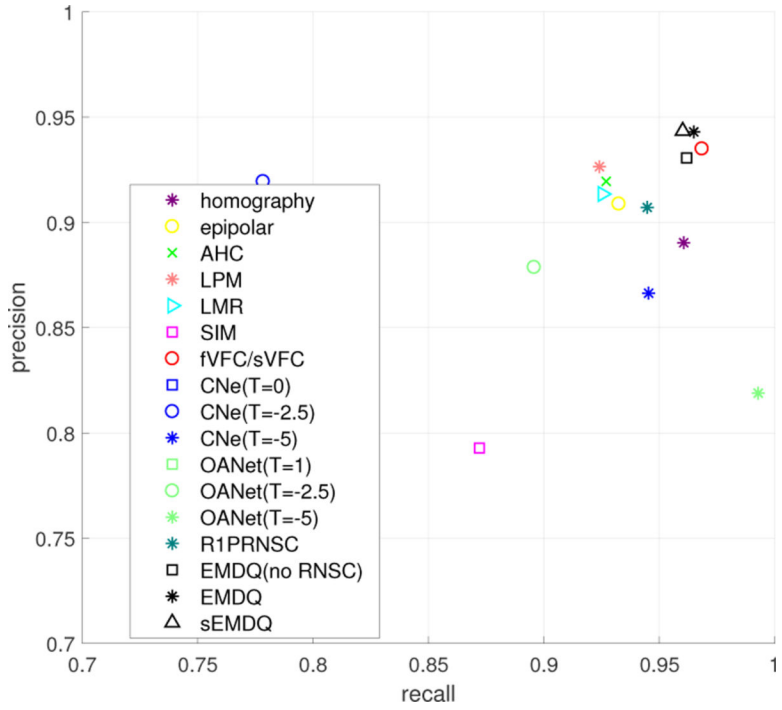


Fig. 7.

Overall precision and recall rates of the DTU experiments in Fig. 5.

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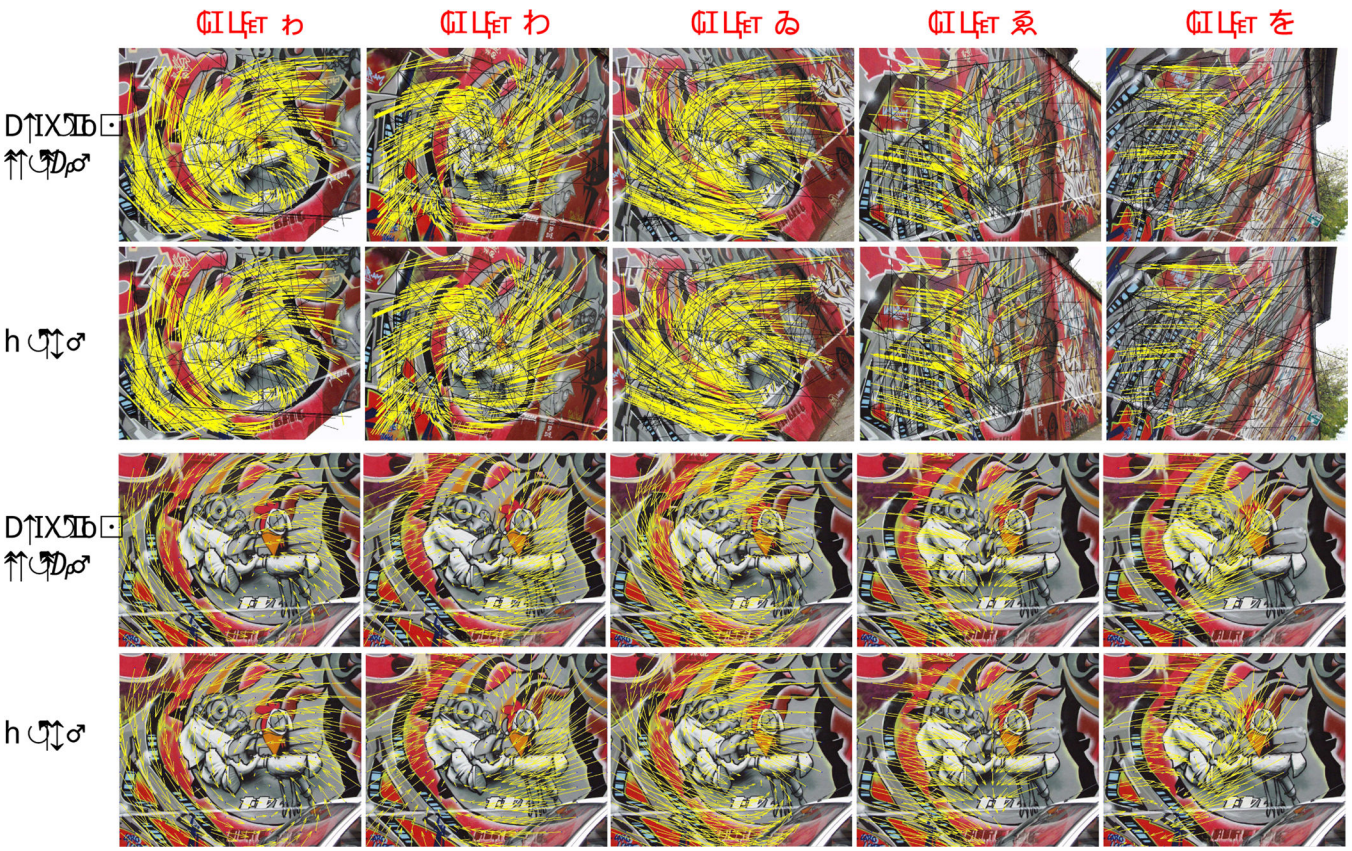


Fig. 8.

Experiments on the Graffiti data from the VGG dataset with AffNet [11] matches, the object is rigid and planer and the ground-truth homography matrices are known. We detected feature points using the AffNet and then matched image 1 with other images using brute-force matching in the descriptor space.

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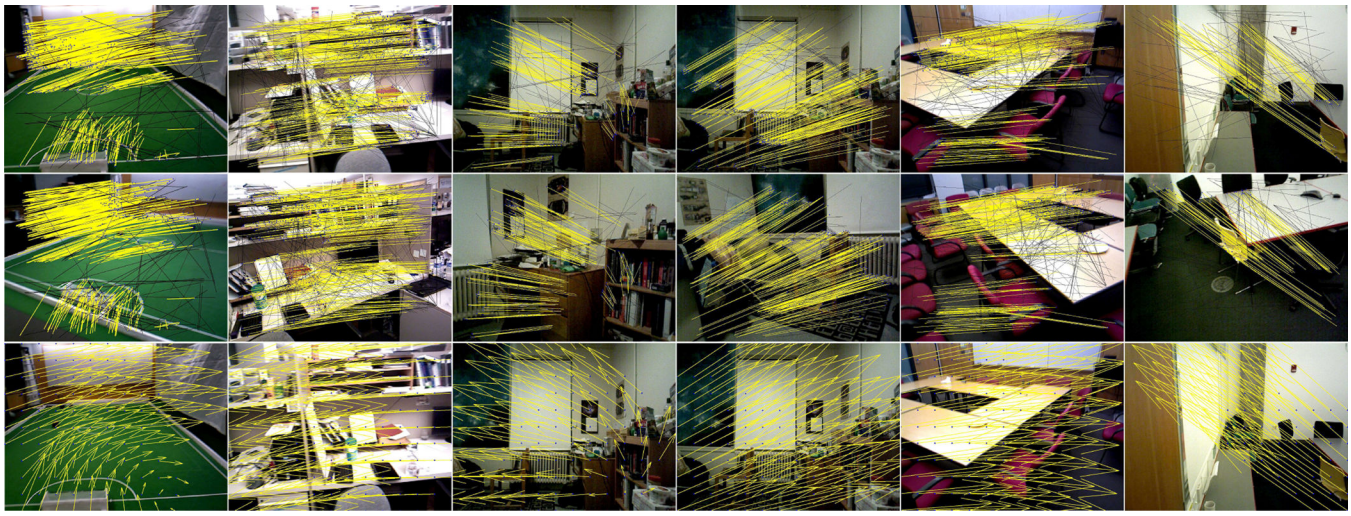


Fig. 9.
Results of EMDQ on the sun3D data with AffNet [11] matches.

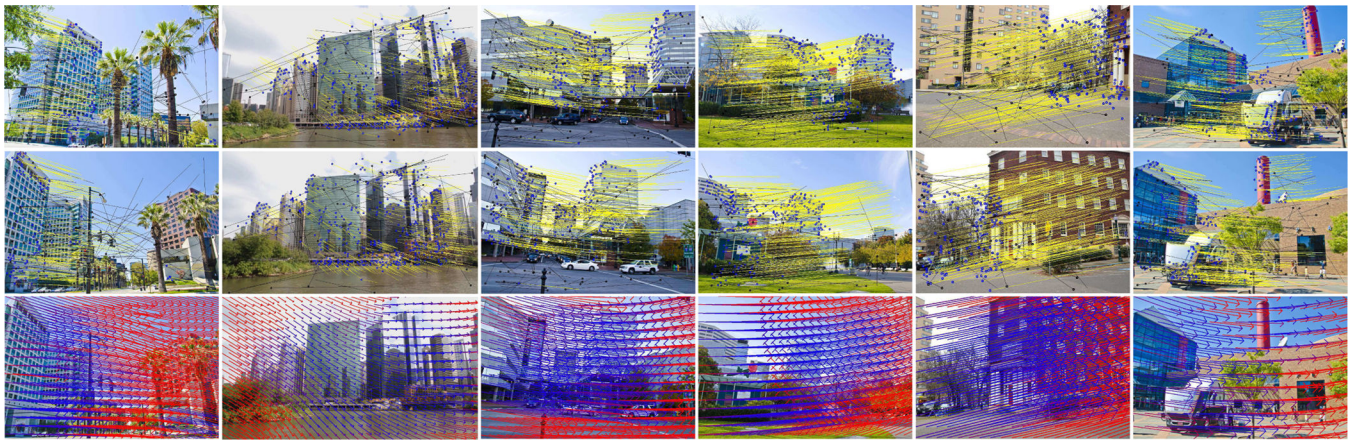


Fig. 10.

Results of qualitative experiments on natural images with SURF matches. The colored arrows of the third row are the deformation field with uncertainties (blue to red: low to high uncertainty) The uncertainties were determined by $\sigma^2 = \exp(0.005d^2)$, where σ^2 is the uncertainty of the deformation field at a pixel, and d is the distance between the pixel and the closest feature inliers.

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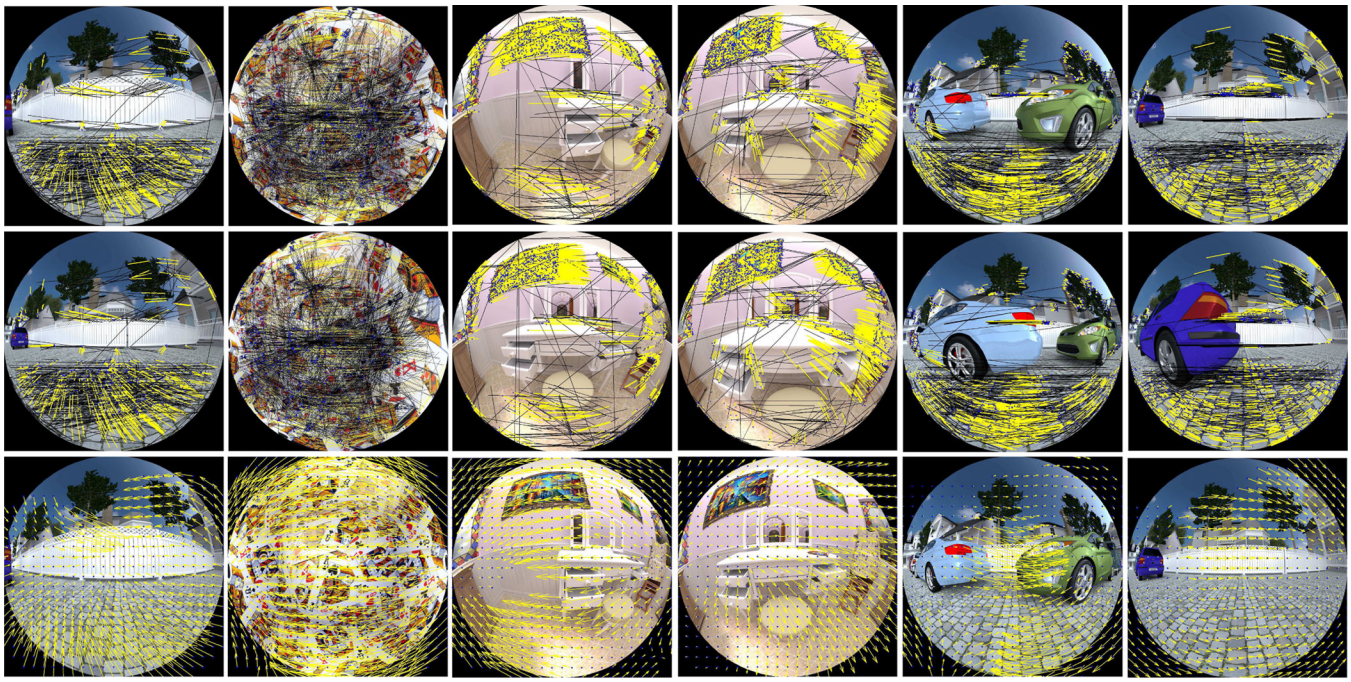


Fig. 11.

Results of EMDQ on fisheye images with AffNet [11] matches. This experiment aims to show the ability of EMDQ to handle image distortion or un-calibrated images.

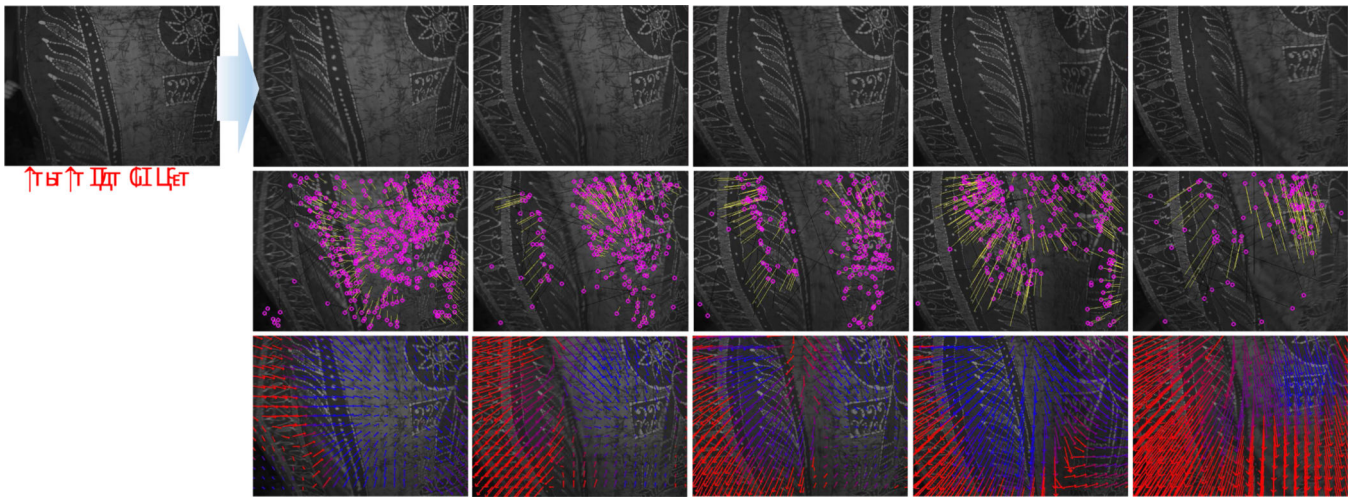


Fig. 12.

Results of qualitative experiments on the non-rigid Mandala data, which includes non-rigid deforming objects. First row: the input images, which are matched with the reference image. Second row: the mismatch removal results from SURF matches. The yellow and black lines are identified as inliers and outliers by EMDQ respectively. Third row: The generated deformation field. Blue to red: low to high uncertainty.

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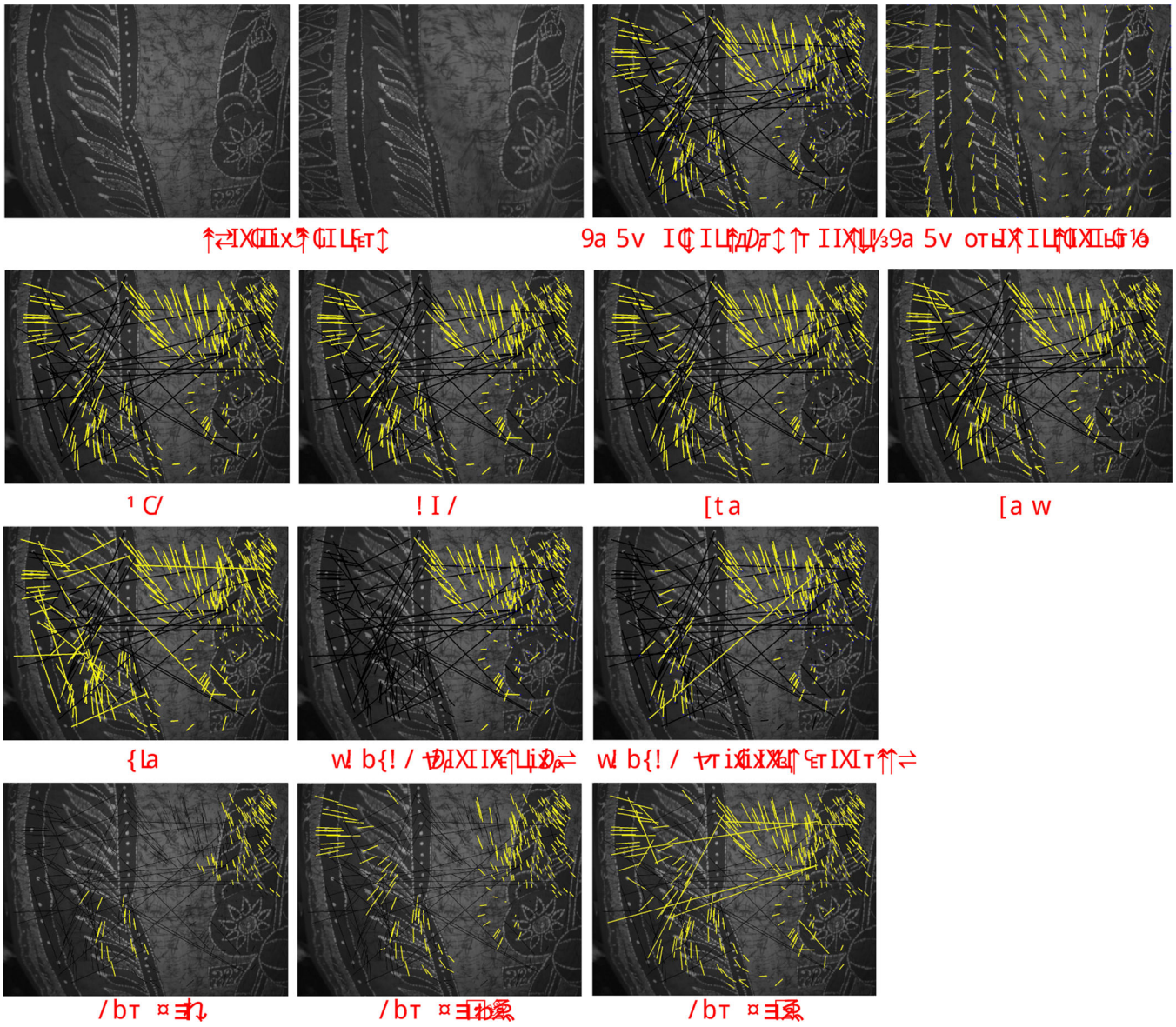


Fig. 13.

Experimental results on the Mandala dataset. Correct and incorrect matches detected by the methods are represented by the black and yellow lines, respectively.

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TABLE I

QUANTITAIVE RESULTS OF THE EXPERIMENTS SHOWN IN FIG. 8.

		RANSAC+ homography	RANSAC+ epipolar	AHC	LPM	LMR	SIM	VFC	CNe (T=-2.5)	OANet (T=-2.5)	R1PRNSC	EMDQ
image2 (correct matches =1487 / total matches =1618)	recall	100.00	95.35	100.00	96.09	95.56	89.50	94.88	74.84	99.86	100.00	99.12
	precision	98.93	99.57	99.79	99.79	99.85	93.93	100.00	98.75	98.34	97.5	99.93
	errors	16	75	3	61	68	242	76	388	27	37	14
	F-score	99.14	98.70	99.83	99.02	98.96	93.01	98.93	92.82	98.64	98.04	99.77
image3 (1043/1178)	recall	100.00	89.45	100.00	90.31	89.93	100.00	97.31	68.74	89.84	98.94	99.42
	precision	98.95	99.46	99.90	99.89	99.89	88.53	100.00	98.08	99.15	95.46	100.00
	errors	11	115	1	102	106	135	28	340	114	60	6
	F-score	99.16	97.28	99.92	97.81	97.72	90.61	99.45	90.37	97.14	96.14	99.88
image4 (728/874)	recall	100.00	87.22	100.00	83.92	77.06	88.18	96.97	71.56	100.00	95.46	94.78
	precision	98.77	99.68	99.86	99.51	99.46	86.63	100.00	97.20	85.05	92.29	99.85
	errors	9	95	1	120	170	185	22	222	128	91	39
	F-score	99.02	96.91	99.89	95.94	94.00	86.94	99.38	90.70	87.67	92.91	98.79
image5 (250/393)	recall	100.00	87.60	100.00	68.00	66.00	90.40	98.00	86.00	99.60	96.80	97.20
	precision	97.65	98.20	99.20	97.70	99.39	69.53	100.00	91.48	97.27	81.48	99.18
	errors	6	35	2	84	86	123	5	55	8	63	9
	F-score	98.11	95.88	99.36	89.85	90.26	72.90	99.59	90.33	97.72	84.14	98.78
image6 (141/282)	recall	100.00	78.72	100.00	63.12	41.13	100.00	98.58	71.63	100.0	100.00	95.03
	precision	95.27	94.87	98.60	93.68	84.05	50.00	98.58	87.06	90.96	69.11	95.71
	errors	7	36	2	58	94	141	4	55	14	63	13
	F-score	96.18	91.13	98.87	85.41	69.54	55.55	98.58	83.47	92.64	73.66	95.58

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